ENG M 401 Summary

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1 Engineering and Business

Question 1.1. (Why should engineers study business?)

1. Engineers take on **managerial roles**, understanding business helps interact with their organization
2. Engineers make **business decisions** (ex. Which projects should have priority, how should things be designed/implemented, etc.)
3. Better business sense will **improve their personal lives**

Definition 1.2. (Types of businesses)

1. **Incorporated Businesses**
   - **Properties**: Business is separate legal entity from owners, ownership held by one or more shareholders, shareholders have limited liability
   - **Classification**:
     - **Private**: Not available for sale on public market
     - **Public**: Shares can be bought and sold freely (most large companies)
     - **For-profit**: Company aims to generate profit and distribute **dividends**
     - **Not-for-profit**: Company aims to advance some form of **social good**
   - **Pros**: Raise capital by issuing stocks/bonds, easy transfer of ownership, trade stocks/shares, limited personal liability
   - **Cons**: Legal restriction on operations, complex taxation, management must meet stockholder expectation
2. **Unincorporated Businesses**
   - **Properties**: No legal distinction between owners and business, can be propreitorship or partnership, owners have unlimited legal liability
   - **Classification**:
     - **Proprietorship**: Business is owned by an individual
       - **Pros**: Owner enjoys all profit, makes own decisions, easy and inexpensive to form, taxed at owners personal tax rate
       - **Cons**: Personal liability, life of owner, cannot issue stocks and bonds
     - **Partnership**: More than one owner
       - **Pros**: More operating capital, more balanced management, easy to borrow from bank
       - **Cons**: Personal liability for partner, bond by each others commitments
3. **Limited Liability Partnerships**
   - **Properties**: Exist for some types of businesses (accounting, law, etc.), partners have full legal liability for own actions, limited for actions of partners

Definition 1.3. (Corporation Structure)

1. **Shareholders**: The owners who provided the money to run the organization
2. **Board of Directions (and Chair of the Board)**: Elected by owners to run the organization
3. **Chief Executive Officer**: most senior officer of the corporation, responsible for operations, finances, and strategy, reports to Board.
4. **President**: responsible for the operation of the entire organization.
5. **Executives**: responsible for different parts of the organization.
6. **Managers**: responsible for line operations.
7. **Staff**: do the work

Definition 1.4. (Engineers in business) **Engineers plan for capital expenditures** (equipment purchases), **make strategic decisions** (equipment/process selection/replacement, product development/expansion, cost reduction, and process/product improvement), and **estimate or forecast**(requirement/investment in a project, product demand/lifespan, selling price, manufacturing costs).

Definition 1.5. (Societies impact on business)

- **Formal Rules**:
    1. Competition or anti-trust laws
    2. Labour Standards and Protections
    3. Environment, Health, and Safety Standards
    4. Professional Associations and legislation

Definition 1.6. (Regulations and Standards Setting) Regulation often reflects **moral standards and social values**. Three broad classes of regulations exist: **protect businesses from businesses**, **protect individuals from businesses**, **protect society at large from businesses**. There are two competing forces for regulation

Definition 1.7. (Money and Values)

1. **What is money?**: A unit of account (can easily be translated into prices), store of value (holds value over time), medium of exchange (widely accepted)
2. **Triple Bottom Line/Three P's**: Growing trend for businesses to consider **people, profit, planet**

Definition 1.8. (Fundamental Principals of Business Management):

1. **Time**: Money today is worth more than money in the future, money has time value, can use it to create future
2. **Difference**: All that counts are differences between alternatives (ignore sunk costs)
3. **Cash Flow**: Cash is oxygen of business. Broad term for the balance of money moving in/out of business over time. More important than long term things (i.e. new product development, retaining good employees, attracting investors)
    - **Operating Cash Flow**: Cash generated from normal business activities (ex. sales to customers)
    - **Investing Cash Flow**: Net cash generated through investment activities (ex. purchase or sale of equipment)
    - **Financing Cash Flow**: Net cash generated thourgh financial activities (ex. selling shares/repaying loans)
4. **Risk**: Additional risk is never taken without addition return. Engineers are in the business of risk management.

2 Financial Statements

Definition 2.1. (Four main types of financial statements)

1. Income Statement - shows performance of the business
2. Balance Sheet - shows snapshot of the current situation of the business
3. Cash flow statement - shows the increases/decreases in cash
4. Retained earnings/shareholder equity statement - shows changes in shareholder ownership

Definition 2.2. (Qualitative Aspects of Financial Reporting)

- **Relevance**: Information can make a difference in the decisions made by users
  - **Predictive Value**: Users can make predictions about future events
  - **Confirmatory Value**: Users can recieve feedback on their predictions
  - **Materiality**: Information is a material omission if it can significantly influence the decisions of users
- **Faithful Representation**: Information represents the economic reality of the entity
  - **Completeness**: All necessary information is presented
  - **Neutrality**: Infomation is unbiased
  - **Free from Error**: Information is formulated with correct processes and supplemented with adequate explanation

Definition 2.3. (Accounting Terms):

- **Bookkeeping**: The process of collecting/classifying/recording raw information that arises from business transactions
- **Journals**: 'Books of original energy'. First-level categorization of transactions, chronological. (ex. cash receipts, cash disbursements, purchases, payroll, sales, general)
- **Ledgers**: 'Books of final account'. Separate journal entries into accounts (ex. cash, accounts recievable, prepaid expenses, cost of sales, bank fees, interst on long term debt). Utilizes sums/running balance of Crs and Dbs. Use **T-format** with Dbs or right and Crs on left.
- **Double Entry System**: Each entry recorder with both a debit (Db) and credit (Cr) entry.
- **Account**: A category or family that we use to categorize information. Five main types:
  - **Assets**: Economic resources controlled by the business, arise from past transaction and future ones from which benefits are obtained
  - **Liabilities**: Obligations owed to creditors/lenders, arise from past transactions (ex. accounts payable, gst payable, loan balance outstanding)
  - **Revenues**: Money going into the business from business activities (ex. sales, interest, etc.)
  - **Expenses**: Money leaving the business as part of business activites (ex. fuel expense, payroll, COGS, travel, benefits)
  - **Equity**: Money provided by the owners to finance the business (ex. common shares, dividends declared, retained earnings)
- **Accounting Equation**: **Debits** inc. LHS, dec. RHS, **Credits** dec. LHS, inc. RHS

$$Assets + Expenses = Liabilities + Equity + Revenues$$
$$Assets = Liabilites + Equity$$

- **Accrual-basis Accounting**: Transactions incurred/earned. Revenues/expenses recognized in same period (**Matching Principle**)
- **Cash-basis Accounting**: Transactions recorded when cash received or paid. NEVER used.
- **Materiality**: The allowable error. Information is material if error or omission would cause users to be misled/draw erroneous conclusions
- **Consistency**: Businesses should keep the same conventions and assumptions thorugh different reporting periods (ex. including labour cost in WIP account)

Definition 2.4. (Accounting Systems)

- **International Financial Reporting Standards (IFRS)**: Used in Canada and around the works, principles-based accounting system
- **Accounting Standards for Private Enterprises (ASPE)**: Used by private enterprises in Canada, principles-based accounting system
- **US Generally Accepted Accounting Practices (US GAAP)**: Used by domestic US companies, rules-based accounting system

3 Income Statements

Definition 3.1. (Income Statement): An **income statement** summarizes a businesses revenues and expenses over a period of time (ex. monthly, quarterly, yearly). Four parts are revenues, variable expenses, fixed expenses, other income. Can be calculated for division/product (do not include taxes or interest), or for comapny (do include taxes and interest).

Definition 3.2. (Financing a Business): There are two main ways to finance a business. **Debt** is borrowing money to pay it back over period of time with interest added (cost of borrowing). **Equity** is exchanging ownership in company for cash. **Debt-to-equity ratio** is used by lenders to determine whether to give you more money.

Definition 3.3. (Income Statement Terms):

- **Gross Revenue**: All money that has been brought in through regular business activities
- **Allowances and Bad Debt**: Main reason why businesses do not collect on gross revenue. Allow company to more faithfully represent true revenue of business/smooth out short-term variation. Are 'contra' asset accounts (hold credit balance)
  - **Bad Debt (Allowances for Doubtful Accounts)**: Allowance for businesses that pay on credit but never pay. Managers should be careful who credit is extended to / how much
  - **Allowances for Discounts**
  - **Allowances for Returns**
  - **Allowances for Warranties**
- **Net Revenue**: Gross Revenue - Allowances
- **COGS/Variable Costs**: Capture costs of making/delivering/buying product. (Raw material to WIP to Finished Items to COGS)
- **Contribution Margin/Margin**: Net Revenue - COGS, CMR = (Net Revenue - COGS)/(Net Revenue)
- **Discounting**: Reduce pricing (ex. more sales, enter new market, react to competition, reduce inventory, incentive for time/volume/payment). Maximum decrease is CMR (if COGS variable and excess capacity). Limited by capacity and margin. Variable/direct.
- **SG&A/Fixed Costs**: Cost of running division, company, items not directly related to manufacturing. Cannot be calculated by product/line. Fixed in theory (for small change in sales), but grows at small rate relative to sales in reality (customer cost). Grows faster than COGS in good times, first to get cut in bad times. Fixed/indirect/overhead
- **Depreciation/Capitalization**: Spread out cost of asset over its useful lifetime. **Non-cash expense** (often shows with SG&A).
- **Breakeven**:
  - **Cash Breakeven**: Operating income, excluding depreciation, becomes positive. CB = SGA/CMR. Business lives in short run.
  - **Book Breakeven**: Operating income, excluding depreciation, becomes positive. BB = (SGA+Dep)/CMR. Business survives in short run and long run at current sales.
- **Operating Income**: Contribution Margin - SG&A - Depreciation
- **Other Income/Non Operating Revenues**: Could mislead users if included in other sections. Includes gains/losses on sale, writedowns, interest, etc.

- **Operating Cash Flow**: Net cash from business activities. Operating Income + Depreciation + Other Income
- **Net Income**: Profit to shareholders.
- **EBIT**: Earnings Before Interest and Taxes (core profitablity of asset base).
- **EBITDA**: Earnings Before Interest, Taxes, Depreciation, and Amortization. EBIT + Depreciation. (EBITDA > EBIT > Net Income, with exceptions)

4 Depreciation

Definition 4.1. (Assets and Depreciation): Are economic resources, arise from past transactions, can produce economic benefit in future. Example is **Property, Plant, and Equipment** which are tangible assets. **Capitalization** is the process of accounting for cost of assets (above **capitalization threshold**) over useful lifetime, associated charge is **depreciation**. Depreciation is an expense (non cash, recorded in SG&A) that reflects need to replace wearing out assets

Definition 4.2. (Accounting Depreciation Terms)

- **Book Value**: Original Asset Cost (Cost basis) - Accumulated depreciation.
- **Writedown**: When we reduce the book value of an asset by choice or due to circumstances (ex. residual when asset taken out of servcie)
- **Impairment**: A one-time book value reduction caused by underlying asset market falling significantly.
- **Gain on Sale of Assets**: Opposite of a writedown. Selling an asset for more than its current book value. Listed under other income. Also called disposal/liquidation of assets.
- **Straight Line Depreciation**: Simplest, most commonly used form of depreciation. $d_t = (B-S)/N$, N = depreciable lifetime.
- **Sum of the Year's Digits**: More accurate model for items that depreciate faster as the start of their life. $d_t = (N-t+1)*(B-S)/SOYD$ where $SOYD = N(N+1)/2$.
- **Declining Balance Depreciation**: Depreciate relative to assets current book value. $d_t = D \times BV_{t-1}$ or $d_t = D \times B \times (1-D)^{t-1}$, $BV_t = B \times (1-D)^t$, and $D = 100/N$. Does not assume salvage value, only depreciate by residual amount if depreciating normally would reduce below salvage.
- **Double Declining Balance**: Same as Declining balance, but with $D = 200/N$. Second most commonly used form of depreciation.
- **Units of Production**: Used when value of asset depends on how much its been used. $d_t = P_t/P_{lifetime} \times (B-S)$.

Definition 4.3. (Tax Depreciation Terms): **Capital Cost Allowance (CCA)** is the tax equivalent of depreciation for taxing purposes. Can be claimed against earnings through CRA (only 50% can be claimed in year asset obtained, **half-year rule**). Assets are pooled into classes, and residual value is called **Unclaimed Capital Costs (UCC)**. UCC is the tax equivalent of book value. **Deferred Taxes** are the difference between tax depreciation and accounting depreciation - it is recorded as charge against income.

1. For each asset, find appropriate CCA rate R
2. For the current year t, find T_t, the total sum of transactions in current year:
   - Purchase positive, sale negative
   - If $T_t > 0$ then $T_t^* = 0.5T_t$ else $T_t^* = T_t$
   - $CCA_t = R \times (UCC_{t-1} + T_t^*)$
   - $UCC_t = UCC_{t-1} + \mathbf{T_t} - CCA_t$ **(Be extremely careful here, USE FULL ASSET COSTS)**

5 Appendix/Examples

Example 5.1. (Types of Accounts)

- **Assets**: Cash, Cash Equivalents, WIP, Accounts Receivable, Finished Goods, Accumalated Depreciation (contra), Prepaid Expenses, Raw Goods, Goodwill, Property Plant and Equipment
- **Liabilities**: Accounts Payable, GST Payable, Current Portion of Long-term Debt, Wages Payable, Short Term Credit Line,
- **Revenues**: Sales, Royalty Income, Rent Income, Interest Gained, Gain on Sale of Assets
- **Expenses**: Travel, COGS, Impairment/Writedowns, Rent, Utilities, Bank Fees, Maintenance and Repairs, Depreciation
- **Equity**: Retained Earnings, Common Shares, Preferred Shares

Example 5.2. (Expense Categorization)

- **COGS**: Shipping charges, transportation insurance, temporary/contract staff salaries, supplies for specfic order, import duties
- **SG&A**: Warehouse/Office lease expenses, Parties, Head office salaries, Phone charges, warehouse staff salaries, travel for sales staff, cost of benefits for permanent staff, snow remove/grounds upkeep, utilities
- **Assets**: Equipment (eg. forklift, vehicle, computer hardware), software (intangible asset)